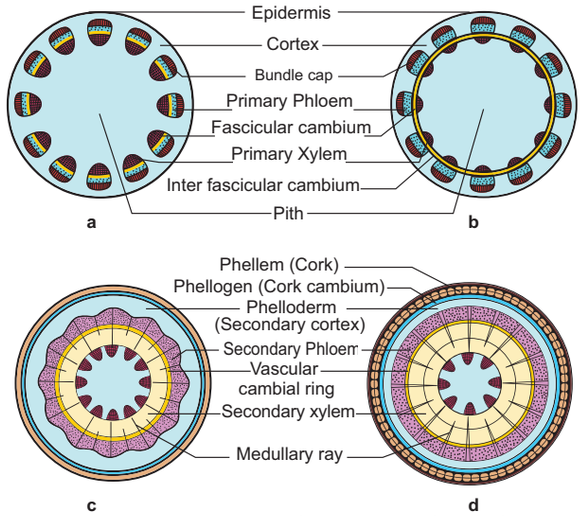


Figure 10.5: Secondary growth in dicot stem (diagrammatic) - stages in transverse section (a–d)

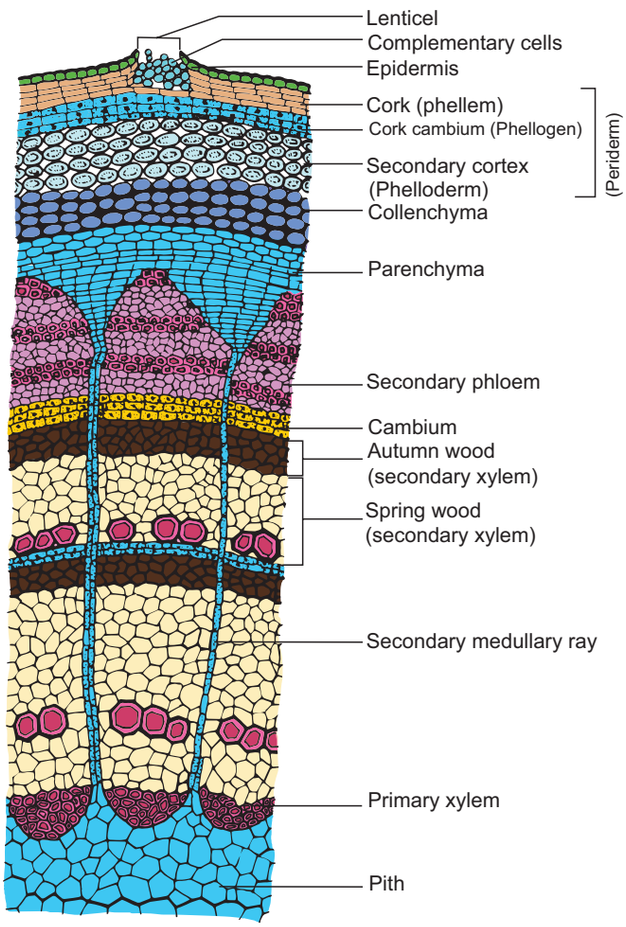


Figure 10.6: Secondary growth in two year old dicot stem – A portion enlarged

Annual Rings

The activity of vascular cambium is under the control of many physiological and environmental factors. In temperate regions, the climatic conditions are not uniform throughout the year. In the spring season, cambium is very active and produces a large number of xylary elements having vessels/tracheids with wide lumen. The wood formed during this season is called **spring wood or early wood**. The tracheary elements are fairly thin walled. In winter, the cambium is less active and forms fewer xylary elements that have narrow vessels/ tracheids and this wood is called **autumn wood or late wood.** The treachery elements are with narrow lumen, very thick walled.



- Usually more distinct annual rings are formed in the regions where climatic variations are sharp.
- Usually more distinct annual rings are formed in temperate plants and not in tropical plants.
- Usually least distinct annual rings are formed in seashore region because the climatic conditions remain same throughout the year.
- Generally annual rings are also less distinct in desert plants.

The spring wood is lighter in colour and has a lower density whereas the autumn wood is darker and has a higher density.

The annual ring denotes the combination of early wood and late wood and the ring becomes evident to our eye due to the high density of late wood. Sometimes annual rings are called **growth rings** but it should be remembered all the growth rings are not annual. In some trees more than one growth ring is formed with in a year due to climatic changes.

Additional growth rings are developed within a year due to adverse natural calamities like drought, frost, defoliation, flood, mechanical injury and biotic factors during the middle of a growing season,which

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